

Deep Learning & AI – Question Bank (5 Marks Each)

Section A: Basics of ML & DL

1. Define Deep Learning. How is it different from Machine Learning? Give two examples.
2. Write a short note on applications of Deep Learning (any 3).
3. Compare ML vs DL in terms of data, performance, and computation needs.
4. Explain why Deep Learning requires large datasets and high computation power.
5. Differentiate between AI, ML, and DL with simple examples.

Section B: Core Concepts

1. What is an Activation Function? Explain any two types with advantages and disadvantages.
2. What is a Loss Function? Explain different types for classification and regression problems.
3. What are Optimizers? Explain any two optimizers with pros and cons.
4. Explain the Vanishing Gradient Problem with a diagram.
5. Write a short note on Gradient Descent and its role in training DL models.

Section C: Artificial Neural Networks (ANN)

1. Explain the working of an Artificial Neural Network (ANN) with a neat diagram.
2. What is Backpropagation? Explain its importance in training ANNs.
3. Explain the role of weights and biases in ANN with an example.

Section D: Convolutional Neural Networks (CNN)

1. What is a Convolution operation? Explain with diagram.
2. Explain different types of Pooling Layers with diagram.
3. Explain the working of a CNN with a neat diagram.
4. What is the role of filters/kernels in CNN? Explain with an example.
5. Write a short note on Fully Connected Neural Networks.

Section E: Recurrent Neural Networks (RNN)

1. Explain the working of an RNN with diagram.
2. Differentiate between RNN and CNN in terms of applications.
3. Explain the vanishing gradient problem in RNNs with respect to long sequences.

Section F: Transformers & Self-Attention

1. Draw the Transformer architecture.
2. Explain the Self-Attention Mechanism in Transformers.
3. Write a short note on Positional Encoding in Transformers.

Section G: Transfer Learning & Applications

1. What is Transfer Learning? Explain with an example.
2. Write a short note on advantages of Transfer Learning.
4. Explain any two real-world applications of Transfer Learning.
5. Explain what is a pre-trained model (e.g., ResNet) and why is there a need of pretrained models

Section H: Object Detection & Segmentation

1. Differentiate between Object Detection and Object Segmentation.
2. Explain different types of Object Segmentation with examples.
3. What is Image Classification? How is it different from Object Detection?
4. Write a short note on Semantic Segmentation.
5. Write a short note on Instance Segmentation with examples.

Section I: AI Agents & RAG

1. What are AI Agents? How are they different from traditional AI models?
2. Explain the RAG (Retrieval Augmented Generation) process with diagram.
3. Explain the role of AI Agents in real-world automation.
4. Explain the components of AI agent (tools , memory , brain)
5. Explain with diagram what is a multi-agent system

Section J: Applications of Deep Learning

1. List five applications of Deep Learning.
2. Explain how AI is used in Healthcare.
3. Write a short note on AI in Finance.
4. Write a short note on AI in Autonomous Vehicles.
5. Write a short note on AI in Education.